Exercise 2.

① SELECT DISTINCT department
FROM students;

department
IT
HR
Finance.

② SELECT department, AVG (age) AS avg_age
FROM students
GROUP BY department;

department	avg_age
IT	20,5
HR	22
Finance	23

③ SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

department	student_count
IT	2
HR	2

④ SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

⑤ SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR') ~~AN~~
AND age >21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

⑥ SELECT department
SUM (credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;

department	total_credits
IT	11

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⑦ SELECT course_id, course_name, department
FROM courses
WHERE credits <> 4 ;

course_id	course_name	department
101	SQL Basics	IT
104	Excel	Finance
105	statistics	HR

⑧ SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

⑨ SELECT
MAX (grade) AS max_grade,
MIN (grade) AS min_grade,
AVG (grade) AS avg_grade
FROM enrollments ;

max_grade	min_grade	avg_grade
90	78	84,6

⑩ SELECT course_id,
COUNT (*) AS enrollment_count
FROM enrollments
GROUP BY course_id ;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

⑪ SELECT department
SUM (salary) AS total_salary
SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finance	70 000	6000

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(12) SELECT department,
AVG(salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG(salary) > 55000;

department	avg_salary
IT	61000
Finance	70 000

(13) SELECT employee_id, name, salary, bonus
(salary + bonus) AS total_compensation
FROM Salaries
WHERE (salary + bonus) > 60 000;

employee_id	name	salary	bonus
3	Spike	70000	6000
4	Tyke	62000	5500

(14) SELECT department
SUM (budget) AS total_budget
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80000

(15) SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120 000
~~AND department <> 'Marketing'~~;
AND NOT department = 'Marketing';

project_id	project_name	department
1	AI App	IT
2	Payroll System	Finance
3	HR Portal	HR

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